

title

Ejercicio 1. Sea $f \in \mathcal{S}$. Demuestra que

$$\forall z, w \in \mathbb{D} : \left| \frac{f(z) - f(w)}{z - w} \right|^2 \leq \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2}.$$

Solución: Como $f \in \mathcal{S}$, por el teorema de Schwarz-Pick, tenemos que

$$\left| \frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)} \right| \leq \left| \frac{z - w}{1 - \overline{w}z} \right| \implies 1 - \left| \frac{z - w}{1 - \overline{w}z} \right|^2 \leq 1 - \left| \frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)} \right|^2.$$

Por el ~~ejer-identidad-auxiliar-disco-unidad~~/ejercicio 1, esta desigualdad se traduce a

$$\begin{aligned} \frac{(1 - |z|^2)(1 - |w|^2)}{|1 - \overline{w}z|^2} &\leq \frac{(1 - |f(z)|^2)(1 - |f(w)|^2)}{\left| 1 - \overline{f(w)}f(z) \right|^2} \\ \implies \left| \frac{1 - \overline{f(w)}f(z)}{1 - \overline{w}z} \right|^2 &\leq \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2}. \end{aligned}$$

Finalmente, observamos que:

$$\begin{aligned} \left| \frac{f(z) - f(w)}{z - w} \right|^2 &= \left| \frac{f(z) - f(w)}{1 - \overline{f(w)}f(z)} \right|^2 \cdot \left| \frac{1 - \overline{f(w)}f(z)}{1 - \overline{w}z} \right|^2 \cdot \left| \frac{1 - \overline{w}z}{z - w} \right|^2 \\ &\leq 1 \cdot \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2} \cdot 1 = \frac{1 - |f(z)|^2}{1 - |z|^2} \cdot \frac{1 - |f(w)|^2}{1 - |w|^2}. \quad \blacksquare \end{aligned}$$